Homework 3
MAC Sub-Layer

Release Date 1404/2/26
Deadline 1404/3/7

Simulation Problems

The **Medium Access Control (MAC)** layer is a crucial sublayer of the data link layer (Layer 2) in the OSI model, responsible for controlling how devices access and transmit data over a shared communication medium. MAC protocols ensure efficient, fair, and collision-free communication in both wired (Ethernet) and wireless (Wi-Fi, CSMA/CA) networks.

In this assignment, we explore MAC layer behavior using the INET Framework and Simulate Switching and other concepts related to MAC layer.

1 Ethernet

In this section, we examine two fundamental networking devices: the **Ethernet Hub** and the **Ethernet Switch**, along with their interaction with the **CSMA/CD** (Carrier Sense Multiple Access with Collision Detection) protocol. While both devices operate at the MAC layer, they differ significantly in how they handle frame forwarding and collision management, directly impacting network performance.

1.1 Switch and Hub

Switches and hubs are two of the most widely used network equapments in many small area networks. Implement the network shown in Figure 1. All the nodes in this network are `EthernetHost` and the hubs are `EtherntHub`. The links are `Eth10M` with a length of 10 meters.

Tasks:

a) Explain the key differences between an Ethernet Switch and an Ethernet Hub and explain their applications. Give a real world example of use cases for each of them

b) Add the following configurations to each of the network nodes in your `.ini` file:

- `sendInterval` = exponential(50ms)
- `respLength` = intWithUnit(uniform(10B, 30B))
- `reqLength` = intWithUnit(uniform(800B, 1200B))

All nodes are working in full duplex mode. Set the `sim-time-limit` to 1s and run the simulation with the following frames:

- host1 → host4
- host3 → server2

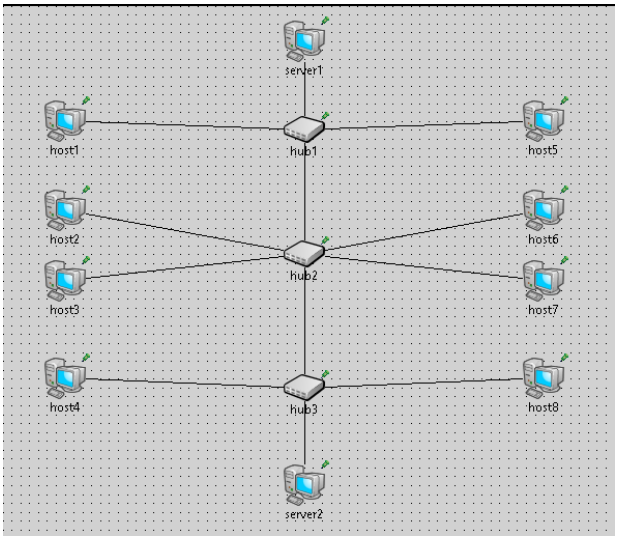


Figure 1: Sample Network using Ethernet Hubs.

- host8 → server1

Calculate the total number of collisions reported by nodes in this scenario. Also calculate the mean **rx channel utilization** of nodes.

c) Now change the `sendInterval` parameter to `exponential(10ms)` and Repeat the previous part. Justify the differences you see in the results.

d) Replace all `EtherntHubs` with `EtherntSwitch` as shown in Figure 2 and Repeat part (b) and (c). Explain the results and differences with previous parts.

e) Provide screenshots of the flow of frames in Hub and Switch networks and Explain the differences between them.

1.2 CSMA/CD

This section investigates CSMA/CD behavior using the topology shown in Figure 2. Configure the simulation with the following parameters:

- Odd-numbered nodes: `csmacdSupport = false`
- Even-numbered nodes and Servers: `csmacdSupport = true`

The traffic pattern is as follows:

- Odd-numbered hosts send requests to `server1`
- Even-numbered hosts send requests to `server2`

Report **rx channel idle**, **rx channel utilization**, and **rx channel collision** for all nodes.

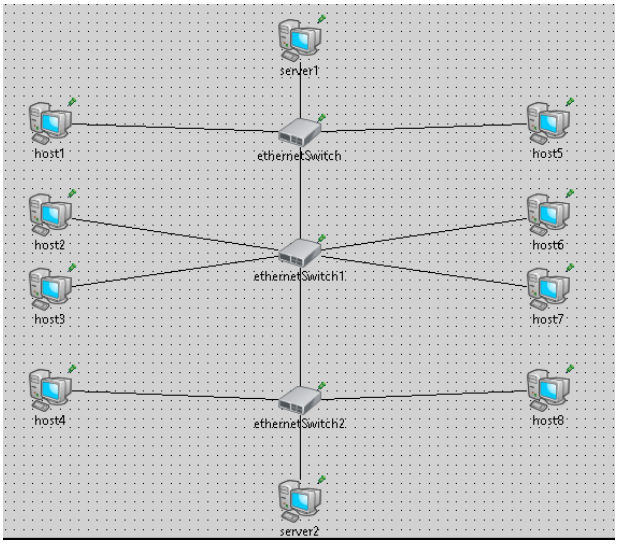


Figure 2: Sample Network using Ethernet Switches.

1.3 MAC Address

Configure the MAC addresses in the NED file according to the following scheme (12 hexadecimal characters in IEEE 802 format):

- Server 1: `00-00-00-00-01-00`
- Server 2: `00-00-00-00-02-00`
- hostX: `00-00-00-00-00-0X` (X = 1, . . . , 8)

Run the simulation using the traffic pattern in section 1.1 and do the following:

a) Display the assigned MAC addresses during simulation initialization.

b) Capture and report the switch's MAC address table.

2 VLAN

Virtual Local Area Networks (VLANs) represent a fundamental network segmentation technology that enables logical grouping of devices regardless of their physical location. This section focuses on implementing and analyzing VLANs using the **INET Framework** in **OMNeT++**, demonstrating their key advantages in modern network architectures.

a) Based on what you learned in class, describe what is a VLAN? When we use them and what are its advantages?

b) All links are `Eth10M` and 10M length. All nodes are `StandardHost`. In the `.ini` file, set VLAN assignments:

- VLAN 10: host1, host4, host7

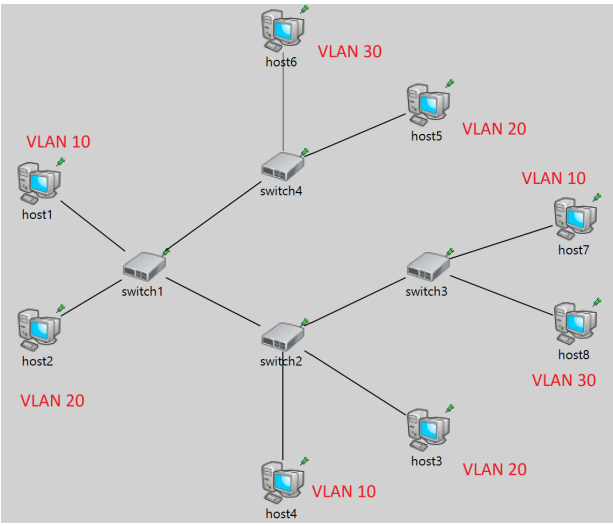


Figure 3: VLAN Network.

- VLAN 20: host2, host3, host5
- VLAN 30: host6, host8

c) Set up ping tests using `Ipv4NetworkConfigurator` in NED and `PingApp` in `.ini`:

- host1 → host4 and host1 → host7
- host8 → host6

d) Analyze the packet flow for both scenarios. Document:

- Switch operations upon packet receipt
- VLAN tagging/forwarding behavior

e) For host1 → host7 communication:

- Trace all nodes/switches receiving ”arpReq” and ”ping”
- Explain ARP’s role in address resolution and why it precedes ping

Submission

Submit all simulation files (.ini, .ned) along with your report, which should describe your simulation steps and answer all assigned questions. Package everything in a ZIP file named **DN_HW3_studentID.zip**, replacing `StudentID` with your actual student ID.